

Manual
Multichannel Switch
(optional)

Scope of delivery

Singlemode Switch

1 x N multichannel

- Optical switch unit
- N* APC/APC single-mode fibers (complete*)
- 1* APC/PC single-mode fiber
- N* collimators (complete*)
- Communication cable 5-pin TTL
- Power supply 5V (Multichannel switches without 5 V socket are powered via the communication cable and do not include an additional power supply)

*Only if the complete version is ordered

Multimode Switch

1 x N multichannel

- Optical switch unit
- N* PC/PC multi-mode fibers (complete*)
- 1* PC/PC multi-mode fiber
- N* collimators (complete*)
- Communication cable 5-pin TTL
- Power supply 5V (Multichannel switches without 5 V socket are powered via the communication cable and do not include an additional power supply)

*Only if the complete version is ordered

Photonic Crystal Switch

1 x N multichannel

- Optical switch unit
- N* APC/APC fused silica input fibers
- N* collimators
- Communication cable 6-pin TTL
- Power supply 5V

HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

Mail: info@highfinesse.com
Web: www.highfinesse.com

Unpacking and recommended way of usage

Please wait 24 h after receiving and unpacking the switch until you connect it to the wavelength meter/power. This makes sure that it has warmed up enough to prevent condensation of water vapor which might damage the switch.

In case the switch is not used for a long time (for example overnight) please disconnect it from the wavemeter and power to avoid degradation of switching mechanics which may occur if the switch is kept fixed at a certain channel for a long time.

In case the switch is not used for a short amount of time it is recommended to just leave the wavemeter in automatic switch mode to avoid long times where the switch is kept fixed to one channel only which would otherwise lead to a faster degradation of the switching mechanics.

Damage threshold of multichannel switches

HighFinesse multichannel switches use micromechanical technology which allows almost unlimited switch cycles, but limits the laser power the switch can endure. To avoid damaging the switch especially with short pulses, please use the wavelength meter in CW mode and use a long exposure time (> 100 msec.). This way you can integrate over several pulses and keep the peak power low. The following limits should not be exceeded:

- **CW lasers:** The limit is 5 mW for all wavelength ranges. For lasers with wavelength < 400 nm the limit is 0.5 mW. HighFinesse wavelength meters usually only need a few µW, this should be more than sufficient, even when transmission losses through the multichannel switch are taken into account.
- **Pulsed lasers:** For the average power of pulsed lasers, the same limits as for the CW case apply. Besides that, a maximum peak power during the pulse of 2 W (0.2W for wavelength < 400 nm) should not be exceeded.

Note: *2 W peak power equals 2 µJ for 1000 ns pulses, 0.2 µJ for 100 ns pulses, and 0.02 µJ for 10 ns pulses! For measurements of short pulses please use long exposure times and integrate over several pulse trains to get the required µJ/sec for a good signal. The average power of such a pulsed laser depends on the duty cycle D and is given by D * 2 W.*

Note: *Fiber connectors can be damaged by too much laser power. Be especially careful when using pulsed (and/or ultraviolet) lasers and calculate the peak power to minimize the risk of reaching the damage threshold of the fiber used.*

HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

Mail: info@highfinesse.com
Web: www.highfinesse.com

Wavelength range

HighFinesse multichannel switches are available as a multimode switch (MM), as a photonic crystal fiber switch (PCS), and with the following singlemode (SM) switch selections: 405 nm, 630 nm, 780 nm, 980 nm, and 1300 nm.

> ***Important:*** *Do not* ***use*** *radiation outside the specified range. The internal optical parts as well as internal fibers may be damaged.*

Model	Wavelength range
SM 405	370 – 650 nm
SM 630	600 – 850 nm
SM 780	750 – 1020 nm
SM 980	950 – 1300 nm
SM 1300	1250 – 2000 nm
Photonic Crystal	390-2000 nm
MM	200-2250 nm

**Switches optimized for Standard and IR range instruments available.*

HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

Mail: info@highfinesse.com
Web: www.highfinesse.com

Electrical connection of a multichannel switch

If the switch has a 5 V socket, please connect the power supply included in the scope of delivery. If the switch does not have a 5 V socket it is powered via the communication cable. Depending on the wavelength meter model, different connection options are possible:

Singlemode: please connect wavelength meter and switch with the grey 5-pin cable.

Multimode: please connect wavelength meter and switch with the grey 5-pin cable.

Photonic Crystal: please connect wavelength meter and switch with the black 6-pin cable.

Note: *There is no need to install separate drivers. The switches are entirely controlled and driven by the HighFinesse wavelength meter software provided on the USB flash drive.*

Older models:

9-pin COM-port: First connect the 5 V power supply to the switch. Then connect the RS232-cable to switch and computer. Be sure to select the correct COM-port in the switch-setting of the HighFinesse software. (Info: Some models have been supplied with an RS232-to-USB adapter. FTDI drivers are usually automatically installed by Windows.)

USB: Connect the USB cable to switch and computer. Check if the FTDI driver is installed automatically by the Wavemeter software. If not, please contact our support.

HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

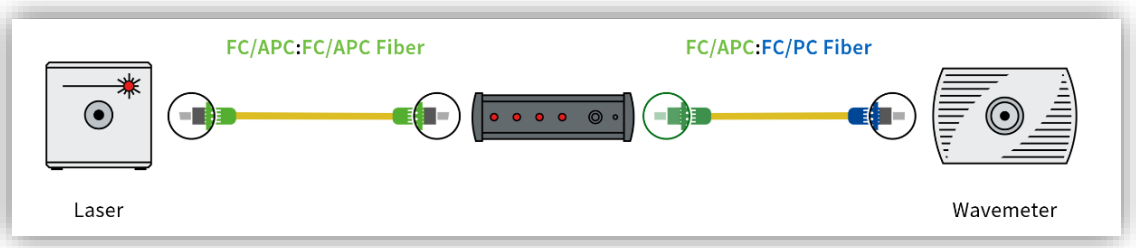
Mail: info@highfinesse.com
Web: www.highfinesse.com

Fiber input and outputs of the multichannel switch

Note: *Ferrule color code: blue for SM FC/PC; green for SM FC/APC; black for MM FC/PC.*

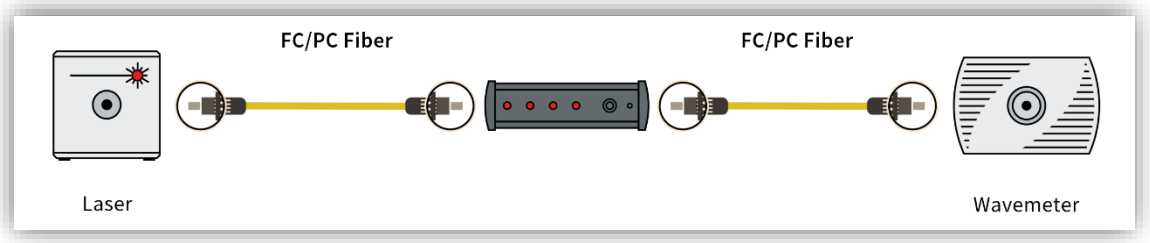
Important: *The use of a FC/PC fiber connector at the wavelength meter is important for correct operation. By pointing a free beam laser signal into the wavelength meter or by using a FC/APC connector the calculated wavelength data will not be reliable.*

SM switch: HighFinesse provides a FC/APC:FC/PC fiber that must be used between wavelength meter and switch. The FC/APC termination is connected to the "common" port of the switch whereas the FC/PC termination is connected to the input port of the wavemeter.



Note: *The fibers provided with the switch are optimized for high transmission. Use analogous SM fibers as provided and used in the switch to ensure a high transmission*

MM switch: Use the FC/PC:FC/PC patchcords to connect the collimators for free beam lasers to the input ports of the switch.



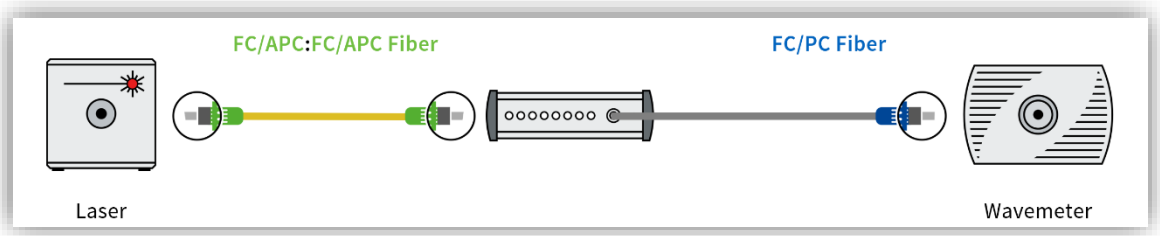
HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

Mail: info@highfinesse.com
Web: www.highfinesse.com

Photonic Crystal switch: These devices have a permanently attached connection fiber that is used to connect the switch to the wavelength meter. This fiber has an FC/PC connector on the wavemeter side. For protection, this fiber is surrounded by a metal jacket.

The inputs of the switch for the fibers from the laser side are all equipped with FC/APC connectors.



Note: *The pure silica PCS input fibers provided with the switch for connecting to the input ports are optimized for transmission over the whole wavelength range of the wavelength meter. These fibers are not single mode for wavelengths lower than 1250 nm. However,* the optics in the photonic crystal switch will ensure that the output is always singlemode. *It is recommended by HighFinesse to use theses provided fibers. Alternatively, existing SM fibers with FC/APC connectors can be used as input fibers to the photonic crystal switch.*

HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

Mail: info@highfinesse.com
Web: www.highfinesse.com

Facilitating Fiber coupling

Coupling into a multimode fiber: Set the collimator into the beam of the laser to be measured and plug in a multimode fiber to the collimator's mount. Vary the height and the adjustment screws of the stand until there is a maximum emission in the optical fiber output.

> ***Note:*** *Multimode quick-coupling can be helpful as a first step before single-mode coupling.*

Coupling into a singlemode fiber/photonic crystal switch input fiber: Plug in an FC/APC:FC/APC SM fiber/photonic crystal switch input fiber and repeat the procedure described above

> ***Note:*** Use the FC/APC:FC/APC patchcords to connect the collimators for free beam lasers to the input ports of the switch. In case you have a fiber coupled laser in most cases a FC/APC connector will be available at the laser. Please confirm with the laser manual before connecting the fiber.

Maintenance

For optimal operation please clean the fiber inputs of the laser patch cord and the switch input at least every fifth fiber change using the fiber cleaning tool provided with the instrument. In case contamination of fiber tips with grease is observed it has to be removed using an appropriate towel and a little bit of 70% ethanol in water. In cases of strong staining, polishing may be necessary. If so, please contact HighFinesse.

> ***Note:*** *Dust on fiber ends can diminish the transmission significantly and can also cause the wavelength calculation to be incorrect. Heavily increasing laser power to compensate can result in permanent transmission loss, by melting dirt onto the fiber end or even damaging it.*

> ***Note:*** *Do not apply force when fixing the fiber connectors to the device to avoid damaging the fiber ends!*

Specifications

General specifications for all models

Available channels:	2,4,8
Return loss	< 40 dB
Crosstalk	< –55 dB
Supply voltage	5 V
Power consumption	450 mW
Operating temperature	0 – 60° C
Storage temperature	–40° – 80° C
Operating Relative Humidity	35 – 55 %
Standard interface	TTL
Optional interface	USB, RS232

Model-specific specification

	Photonic Crystal Switch	Singlemode (SM) Switch	Multimode (MM) Switch
Guaranteed transmission	15 % (@ 632)	20 % (within the specified wave-length range)	50 %
Switching time	typ. 12 ms	typ. 12 ms	typ. 25 ms
Dimensions, 2 and 4 channels	165 × 242 (+fiber) × 43 mm	175 × 110 × 48 mm	175 × 110 × 48 mm
Dimensions, 8 channels	222 × 200 (+fiber) × 43 mm	225 × 110 × 48 mm	225 × 110 × 48 mm

Wavelength ranges

Model	Wavelength range
SM 405	370 – 650 nm
SM 630	600 – 850 nm
SM 780	750 – 1020 nm
SM 980	950 – 1300 nm
SM 1300	1250 – 2000 nm
Photonic Crystal	390-2000 nm
MM	200-2250 nm

HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 - 0
Fax: +49 (0)7071 539 18 - 99
LinkedIn: /company/highfinesse/

Mail: info@highfinesse.com
Web: www.highfinesse.com

Warranty Terms

HighFinesse offers highest quality standards and is implementing strict quality ensuring methods to satisfy customer needs. In case some malfunction should happen, the following warranty terms apply:

12-month warranty

HighFinesse undertakes to replace or repair, at its option, any goods supplied by HighFinesse (save as provided in (f) below) if a defect in material or workmanship arises under conditions of normal and proper use and maintenance (fair wear and tear excepted), provided that all of the following are true:

a) The goods were operated and maintained in accordance with HighFinesse's operating instructions.
b) The claim is first notified promptly in writing to HighFinesse.
c) The defect occurs within 12 months from the date of Take-Over of the goods by the buyer.
d) The goods have not been repaired or modified by anyone other than HighFinesse or at the HighFinesse's direction.
e) In the case of goods not of its own manufacture, HighFinesses's responsibility shall be limited to the passing on to the buyer the benefit of any guarantee or warranty given to HighFinesse by the manufacturer of such goods.
f) Fibers, collimators, USB cables, power supplies, and other accessorial items are themselves excluded from the warranty provisions.

HighFinesse GmbH reserves the right to withdraw warranty terms from buyers who fail to meet the agreed payment terms.

Return of goods

During the period of warranty, the cost of returning the goods to HighFinesse will be met by the buyer or local distributor, with HighFinesse paying the return cost from the factory to buyer or distributor.

Warranty Extension

The option to extend the warranty of wavelength meters beyond the standard 12 months is only available at the time of purchase.

HighFinesse Terms and Conditions of Sale and Supply

1. General
I. These "terms of conditions" apply for all offers, order confirmations, deliveries, and ser-vices from/by HighFinesse GmbH (in the follow-ing "the vendor") as an integral part of the con-tractual relationship. Changes, differing condi-tions and/or supplements to the agreement are binding only in the case of a written confirma-tion from the vendor.
II. The customer's conditions of purchase are invalid even if they have not been explicitly dissented.
III. The terms of conditions come into force with receipt of the order confirmation at the latest.
IV. The information provided from the custom-er in connection with the order is not considered confidential unless explicitly agreed otherwise in writing or specified by the customer in writing.
V. In doubt, in each case, valid INCOTERMS will be basis of any interpretation of International Commercial Terms.

2. Offer and Price, Right to Documentation
I. Unless otherwise agreed in writing, the ven-dors' offers are subject to change and non-binding. This remains subject to the goods being unsold.
II. The scope of the vendor's obligation to deliv-er will be determined solely by the written or-der confirmation. The order remains subject to technical changes, which do not affect the func-tion of the ordered items, prior to delivery.
III. Details given in catalogues and prospectuses, as well in documents, diagrams, drawings, weights and measures given in offers and or-ders are non-binding and for information only, as long as not specifically mentioned in the or-der confirmation.
IV. The vendor retains ownership and copyright of/to any diagrams and other documents. They may not be sent, copied, or made available to third parties and bestow no right to repli-cate/reverse engineer individual parts.
V. The prices relate to the scope of services and products described in the order confirmation.
VI. The stated prices are for goods delivered from Tübingen and include/exclude, as agreed upon, customs duties, with the addition of VAT based on the rate at the time of billing. Prices are calculated depending on the type and ex-tent of the scope of work as agreed upon in the order confirmation and are subject to change if the customer requests additions or changes. Prices are based upon the current prices of the vendor's suppliers, wages, currency parities and customs and import tariffs at the time of the order confirmation. Fixed prices require prior written agreement. The vendor also reserves the right to alter prices subject to changes in quantities.

3. Payment Conditions
I. The vendor's invoice is to be paid within 30 days after receipt without deductions. The cus-tomer is responsible for paying any transaction fees.
II. Subject to proof of greater losses/damage, the vendor may charge default interest of 8% above the basic rate.

III. In the event of default/delay with a pay-ment, the vendor's outstanding claims become due. In addition, the vendor has the right to put any outstanding work on hold. Agreed delivery times are extended accordingly.
IV. Counterclaims by the customer are effective only if recognized by the vendor or legally certi-fied. A lien may only be exercised when it is based on the same contractual obligations.

4. Delivery

I. All stated delivery times are, unless otherwise agreed, approximate and subject to current circumstances such as production capacity and market situation. If the delivery time is deter-mined by a specified period, this period begins after the conclusion of the contract as soon as all associated requisites on the part of the cus-tomer (e.g., transfer of documents, required data etc.) are fulfilled, licenses or permissions obtained and agreed numbers are specified.
II. Subsequent desired changes from the cus-tomer, subject to consideration by the vendor, lead to an appropriate shift of delivery times and deadlines.
III. Timely delivered devices, received by the customer, which have only inconsequential imperfections, are considered to fulfill the de-livery time requirements. Part deliveries are permissible. Delays to delivery due to unpredictable events beyond the vendor's control such as Acts of God, strikes, lockouts, black-outs, shortages of important tools/raw materi-als, sabotage or delays to deliveries of neces-sary tools/raw materials, bring an appropriate delay to the delivery deadline into effect. Rights to cancellation and further claims from the cus-tomer are not effective in such cases. The same applies when, due to unpredictable events, the vendor is already in default. A right to cancella-tion due to delivery delays exists only under the stipulations of clause 9 and claims to damages only in accordance with clause 8.
IV. When the contract relates to the supply of non-fungible goods which are to be manufac-tured or created, the customer only has a uni-lateral right of termination until transfer of the risk (see clause 5) if an important ground exists. If the vendor is not liable for said ground, the vendor's claim to remuneration shall amount to 30% of the order value, unless the statutory claims are higher or lower, the latter to be proven by the customer. If the contract relates to other articles, the subsequent termination of contract will be subject to the vendor's approval and the vendor will charge 30% of the order value, unless agreed otherwise.

5. Transfer of Risk/Perils (under German Civil Law, the "Gefahrübergang", or Transfer of Risk, Refers to the Point at Which Responsibly for Goods, including Loss or Damage, Passes from the Vendor to the Customer)

I. Risk is transferred to the customer upon dispatch of goods. This also applies to part deliveries and if the vendor has taken on the transport and/or installation costs.
II. If the delivery is delayed due to circumstances for which the vendor is not liable, risk is transferred to the customer upon written notice of readiness to dispatch.

6. Responsibility for Defects

I. The vendor guarantees that the product and its performance are defect free at the point of risk transfer. Unless agreed otherwise, the agreed quality corresponds to the vendor's specifications, as amended and published at the time of the order confirmation. The statutory inspection and complaint obligations must be observed.
II. If the goods or services are not of the agreed quality at the time of risk transfer, the vendor will rectify the situation (provide supplementary performance) in accordance with section 439 of the German Civil Code, unless otherwise stipulated in the following provisions.
III. Supplementary performance may be provided by the rectification of defects or, at the vendor's discretion, by new delivery. The vendor may rectify a certain defect several times and, again at the vendor's discretion, may proceed from a rectification of defects to new delivery. Transport costs due to the goods being brought to a location other than the place of delivery are borne by the customer.
IV. The customer may set an appropriate period of at least two weeks for the vendor to affect the new delivery and, if the new delivery does not occur during such period, may demand reduction of the price or, unless the defect is insignificant, withdraw from the contract after expiry of the period. Damages for defects of quality may be claimed only subject to clause 8 below.
V. Rights and claims based on defects of quality expire 24 months after delivery, except in the case of willful intent. However, damage claims based on defects of quality also expire after the statutory period even in the case of loss of life, bodily harm, detriment to health or gross negli-gence.
VI. The vendor is not liable for regular wear and tear (except during 12 months after delivery), customer-procured material, or adjustments made by the customer, damage resulting from improper storage, installation or operation, insufficient maintenance, or for damage result-ing from a modification or repair not authorized by the vendor in writing.
VII. With respect to goods or services which the vendor receives from a third party for resale to the customer, the vendor assigns all warranty rights against such a third party to the custom-er. Claims can be advanced against the vendor only if they have first been asserted against the third party without success.

7. Liability for Property Rights Infringements

I. The vendor warrants upon risk transfer that no third-party patents or other proprietary rights exist which may be claimed with respect to the goods or services if these are used as intended. Clauses 6.2 to 6.5 and 6.7 shall apply accordingly.
II. The vendor's liability will be excluded if a third-party patent or proprietary right is in-fringed because the vendor has adhered to a design provided by the customer or has com-plied with an instruction given by the customer, or because the goods are used for a purpose, in a country, or in connection with other goods or other software, without this having been com-municated to the vendor before execution of the contract.
III. During the period of the vendor's liability, the customer has the obligation to inform the vendor in writing as promptly as possible in the event that a third party claims any patent or other proprietary right or asserts any claims in or out of court with respect to the goods or services. Before recognizing any claim advanced by a third party in or out of court, the customer shall give the vendor the opportunity to com-ment. Upon request, the vendor shall be given the authority to handle the negotiations or legal dispute with such a third party at the vendor's own cost and responsibility. The customer shall be liable to the vendor

for any damage sus-tained as a result of a culpable violation of said obligations.

8. Damage Claims

I. We are only responsible for damages caused by intent or gross negligence or based on the provisions of the German Product Liability Act.
In case of slight/ordinary negligence we are only responsible for damages
a) due to harm done to a person's life, body or health,
b) due to the culpable breach of a material con-tractual duty or to the breach of a duty the dis-charge of which allows the proper performance of the agreement in the first place and on the fulfilment of which the Customer can regularly rely on and if we are responsible for such breach of duty, we will be liable in accordance with the statutory provisions. In such event, however, liability shall be limited to foreseeable and typical damage.
II. Except in case of intentional damage by the vendor's personnel or causation of damage due to gross negligence by the vendor's statutory representatives or executive employees, the vendor's liability shall be limited, in each case, in terms of amount, to the damage which is typically foreseeable at the time of conclusion of the contract.
III. In case of doubt, the amount of typically foreseeable damage shall be limited to the or-der value, up to a maximum of € 50,000.
IV. Claims to damages which result from loss of life, bodily injury or detriment to health, as well as damage claims under the Product Liability Act remain unaffected.

9. Rescission

A breach by the vendor which is not a defect entitles the customer to rescind only if the ven-dor is responsible for such a breach and if the other statutory requirements are met.

10. Retention of Title

I. The vendor retains title to the goods delivered until full payment of the vendor's claims due under this contract, including ancillary claims and future claims arising from the business relationship with the customer.
II. The customer is entitled to resell the goods delivered subject to the vendor's retention of title in the ordinary course of business. The goods must not be pledged or assigned by way of security. If a third party intends to seize the goods subject to the vendor's retention of title, the customer will advise the third party of the vendor's title and inform the vendor immediately.
III. The customer will assign its claims resulting from any resale to the vendor. The customer is authorized and obliged to collect the claims in the ordinary course of business. At the vendor's request, the customer will name the claims as-signed and the debtors concerned. To secure payment claims, the vendor is entitled to dis-close the assignment of the claim at any time.
IV. If goods to which the vendor holds title are connected or intermixed with other goods to which the vendor holds no title, the vendor becomes proportionate joint owner in relation to the invoice value of the goods delivered to the other goods. Any processing or conversion of the goods by the customer will always be undertaken for the vendor, as manufacturer, without this obliging the vendor in any manner. The vendor becomes co-owner of any pro-cessed or converted goods in the sense of the aforesaid provisions.
V. If the customer is in delay of payment, in-cluding with respect to ancillary or future goods delivered or services provided, or if the custom-er becomes insolvent, the vendor may claim retention of title and retrieve the goods, even by entering the premises of the customer. If the vendor claims retention of title, the vendor will not be deemed to have rescinded the contract, unless the vendor declares otherwise.
VI. If the value of the collateral should exceed the vendor's payment claims by more than 20%, the vendor will release the excess collateral at the request of the customer.

11. Onward Delivery

I. The vendor draws attention to the fact that parts of the product range are covered by Ger-man and/or American export specifications and are not intended for re-export. Before any on-ward sale or application of products drawn from the vendor to areas outside the Federal Republic of Germany, the customer takes it upon themselves to make appropriate enquiries to the relevant authorities or industrial or trade bodies and to conduct themselves lawfully in this respect.
II. The customer agrees to account for the whereabouts of the products if the vendor re-quests this stating justifiable reason to do so.
III. The customer commits to imposing the I. and II. aforementioned obligations to any third-party customer with a view to dissemination. If the customer becomes aware of any breach of these obligations by his third-party customers, he will inform the vendor of this promptly.

12. Miscellaneous

I. The vendor's legal relationship with the cus-tomer is subject to the substantive law of the Federal Republic of Germany, excluding the UN Convention on Contracts for the International Sale of Goods (CISG).
II. The place of fulfillment of the parties' mutual obligations is Tübingen, Germany.
III. If the customer is a fully qualified merchant in the sense of the German Commercial Code or has no general place of jurisdiction in Germany, the vendor may elect that the Landgericht Tübingen (Regional Court) or the Landgericht at the customer's place of business shall have exclusive international, local and subject-matter jurisdiction.
IV. If any provision of these General Terms and Conditions of Sale and Supply is or becomes invalid or unenforceable, the other provisions shall remain in full force and effect. The invalid provision shall be replaced by such provision which is valid and comes closest to the intended economic purpose of the invalid provision.

Contact information and service address

For further information please feel free to contact us.



HighFinesse GmbH
Wöhrdstrasse 4
72072 Tübingen (Germany)

Tel.: +49 (0)7071 539 18 – 0
Fax: +49 (0)7071 539 18 - 99

mail: info@highfinesse.com
web: www.highfinesse.com

LinkedIn: /company/highfinesse/